

Applying Blackwell's Sufficient Condition to OGM

Now let's go back to the optimal Growth model.

The recursive formulation:

$$V(k) = \max_{k' \in \Gamma(k)} \{F(k, k') + \beta V(k')\} \quad \text{Bellman's Equation}$$

denote by $TV(k)$

To establish that T is contraction (and, hence, the Bellman Equation has a unique solution $V(k)$) let's check if it satisfies monotonicity & discounting.

1. Monotonicity:

Let $V(k) \leq W(k)$, $\forall k$.

$$\begin{aligned} (TV)(k) &= \max_{k' \in \Gamma(k)} \{F(k, k') + \beta V(k')\} \leq \\ &\leq \max_{k' \in \Gamma(k)} \{F(k, k') + \beta W(k')\} = TW(k) \end{aligned}$$

2. Discounting:

$$\begin{aligned} (T(V+d))(k) &= \max_{k' \in \Gamma(k)} \{F(k, k') + \beta (V(k') + d)\} \\ &\quad \uparrow \\ &\quad \text{constant } d \geq 0 \\ &= \max_{k' \in \Gamma(k)} \{F(k, k') + \beta V(k') + \beta d\} \\ &\quad \uparrow \\ &\quad \text{independ. of } k' \\ &= (TV)(k) + \beta \cdot d \end{aligned}$$

$\Rightarrow T$ is a contraction of modulus $\beta \in (0, 1)$

$\Rightarrow$ It has a unique fixed point.

Solving Bellman Equation

Now we know that there exists a unique $V(k)$ solving the Bellman Equation:

$$V(k) = \max_{k' \in F(k)} \{ F(k, k') + \beta V(k') \} \quad \text{(RP)} \quad \text{recursive problem}$$

How to find such $V(k)$?

Two main approaches:

- ① Analytical solution - "Guess and Verify"
works for a very limited set of $F(k, k')$
- ② Numerical solution (when $\nexists$ analytical solution)

① Analytical solution - "Guess and verify."

Since V is "generated" by some modification of $F(k, k')$, it often inherits its properties, including the functional form.

In our main model, $F(k, k') = u(f(k) - k')$

$\Rightarrow$ we should expect that $V(\cdot)$ may "inherit the shape" of $u(\cdot)$

Example: $u(c) = \log c$, $f(k) = Rk$

Recall that in the (SP), the solution was $k_{t+1} = \beta R k_t$.

Let's see what we get in (RP) (should be the same)

Guess: $V(k) = A + B \cdot \ln k$

Solve for $k' = g(k)$ given this guess: $v(k')$

RHS of (RP): $\max_{k' \leq Rk} \ln(Rk - k') + \beta [A + B \cdot \ln k']$

FOC $\Rightarrow \frac{1}{Rk - k'} = \beta \cdot B \frac{1}{k'}$

$$k' = \beta B \cdot Rk - \beta B k'$$

$$k' = \boxed{\frac{\beta B}{1 + \beta B} \cdot Rk = g(k)}$$

Verify that the implied $V(k)$ is indeed $A + B \ln(k)$

Plug $g(k) = \frac{\beta B}{1 + \beta B} \cdot Rk$ into the (RHS) of (RP):

$$\Rightarrow V(k) = \log\left(Rk - \frac{\beta B}{1 + \beta B} Rk\right) + \beta \cdot [A + B \cdot \log\left(\frac{\beta B}{1 + \beta B} \cdot Rk\right)]$$

$$= \log\left(\frac{R}{1 + \beta B}\right) + \log(k) + \beta A + \beta B \cdot \log\left(\frac{\beta B}{1 + \beta B} R\right) + \beta B \log k$$

$$= \beta A + \log\left(\frac{R}{1 + \beta B}\right) + \log\left(\frac{\beta B}{1 + \beta B} \cdot R\right) + (1 + \beta B) \log k$$

$$= V(k) \Rightarrow \text{must be equal to } A + B \log k \text{ for every } k$$

$$\Rightarrow \int 1 + \beta B = B$$

$$\int \beta A + \log\left(\frac{R}{1 + \beta B}\right) + \beta B \cdot \log\left(\frac{\beta B}{1 + \beta B} \cdot R\right) = A$$

$$\Rightarrow \boxed{B = \frac{1}{1 - \beta}}$$

$$\Rightarrow g(k) = \frac{\beta / (1 - \beta)}{1 + \frac{\beta}{1 - \beta}} \cdot Rk = \boxed{\beta Rk = g(k)}$$

Plug this B into the second equation to find A .

consistent with what we got in the (SP)!

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$$\beta A + \log((1-\beta)R) + \frac{\beta}{1-\beta} \log(\beta R) = A$$

$$\Rightarrow \boxed{A = \frac{\beta}{(1-\beta)^2} \log(\beta R) + \frac{\log((1-\beta)R)}{1-\beta}}$$

$$\Rightarrow V(k) = \frac{\beta}{(1-\beta)^2} \log(\beta R) + \frac{\log((1-\beta)R)}{1-\beta} + \frac{\log k}{1-\beta}$$

↑

We just verified that this f_n solves (RP).
Since (RP) has a unique solution, we are
done!!!

Using recursive argument to establish properties of $V(k)$

Closed-form solution, however, often does not exist.

Can we establish any properties of the optimal choice?

Normally, one would use the FOCs...

When is the first order approach valid?

$$V(k) = \max_{k' \in \Gamma(k)} \{F(k, k') + \beta V(k')\} \quad (RP)$$

$F(k, k')$: increasing in k
decreasing in k'

$\Rightarrow$ for the "trade-off" to exist, $V(k')$ must be $\uparrow$ in k'
(otherwise choose the smallest k')

Result 1: $V(k)$ - increasing if

- $F(k, k')$ is strictly $\uparrow$ in k
- $F(k)$ is increasing in k

Proof: "recursive" argument. Very useful to establish various properties of the solution.

Idea: show that operator T preserves the desired property:

- suppose V_0 has property (P)

- verify that TV_0 has the same property
 $\Rightarrow T^2 V_0$ has it too $\Rightarrow \dots \Rightarrow T^n V_0$ has property (P)

$\Rightarrow$ /take limit/ $\Rightarrow V = \lim_{n \rightarrow \infty} T^n V_0$ has the weak version of this property

- if want to show that (P) is strict $\Rightarrow$
(e.g. strict monotonicity)

$\Rightarrow$ establish that strong (P) holds for TV_0 even if V_0 satisfies weak (P).

$\Rightarrow \lim_{n \rightarrow \infty} T^n V_0$ satisfies weak (P)

But T applied to $\lim_{n \rightarrow \infty} T^n V_0$ would satisfy strong (P) $\Rightarrow V = TV$ satisfies strong (P)

Apply this approach here:

Let $V_0(k)$ - weakly increasing in $k \Rightarrow$

Show that $(TV_0)(k_1) < (TV_0)(k_2) \quad \forall k_1 < k_2$ feasible.

$$(TV_0)(k_1) = \max_{k' \in \Gamma(k_1)} \{ F(k_1, k') + \beta V(k') \} \leq$$

$$< \left/ \begin{array}{l} F(k_1, k') < F(k_2, k') : F \text{ is strictly increasing in } k \\ \Gamma(k_1) \subseteq \Gamma(k_2) : \Gamma(\cdot) \text{ is increasing in } k \end{array} \right/$$

$$< \max_{k' \in \Gamma(k_2)} \{ F(k_2, k') + \beta V(k') \} = (TV_0)(k_2)$$

$\Rightarrow$ T converts a weakly increasing f.n into a strictly increasing f.n.

$\Rightarrow \lim_{n \rightarrow \infty} T^n V_0$ is weakly increasing

$\Rightarrow T$ applied to this limit f.n is strictly increasing 

Can be verified that $F(\cdot, \cdot), \Gamma(\cdot)$ satisfy the desired properties when $u(\cdot)$ - strictly increasing
 $f(\cdot)$ - strictly increasing